SHASHANK HIRWANI

Full-Stack, GenAI & Mobile Engineer

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PROFILE

Full-Stack, GenAI & Mobile Engineer building **hardware-accelerated AI pipelines** (ROCm/CUDA), **3D digital twins** (Three.js), and **offline-first mobile systems** (React Native/Expo). Track record shipping MoA AI architectures, deterministic AST analysis, Django/Next.js platforms, and Moodle REST integrations.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C, C++, Java, SQL, HTML/CSS

AI & GPU: Fireworks AI, Gemini API, Mixture-of-Agents (MoA), AMD ROCm, NVIDIA CUDA, Prompt Engineering

Frontend & 3D: React 19, Next.js, Vue 3, Three.js (WebGL), Tailwind CSS, Vite

Mobile & Backend: React Native 0.86, Expo SDK 57, Django REST, Node.js, Supabase, PostgreSQL

DevOps & Testing: Git, GitHub, Vercel, Jest, Moodle REST API, JWT Auth, Postman, Swagger/OpenAPI

EXPERIENCE

United Nations OICT — New Delhi
United Nations Volunteer – Software Development — *Mar 2026 – Aug 2026*

- Delivered software development and technical documentation support to **UN India OICT New Delhi.**
- Engineered the **UNIlearn Maharashtra** mobile companion app (GitHub) to complement the UN OICT Delhi web portal; implemented an offline-first architecture using **React Native 0.86** and **Expo SDK 57** with **Moodle REST API 4.x**, encrypted token storage, trilingual UI, and **60+ Jest tests**.

Safaikart (Freelance) — Remote
Android Developer – Expo Go / React Native — *Mar 2026 – Present*

- Architecting a production Android app for laundry, delivery, and home-cleaning services with real-time order tracking and push notifications.
- Accelerated feature velocity **40%+** via **AI-assisted workflows** for code generation, debugging, and scalable state management across 15+ screens.

OUTLIER.AI — Remote
Individual Contractor – Prompt Engineering — *Oct 2024 – Dec 2024*

- Optimized AI and LLM responses by designing prompts that **intentionally generate errors** to enhance model learning and improve reasoning accuracy.
- Reviewed code and AI-generated outputs to identify deviations and implement response enhancement strategies; trained AI models to refine their understanding and generate **accurate, context-aware responses**.

PROJECTS

RadeonShift AI — GitHub · Live Demo
TypeScript, Python, Fireworks AI, AMD ROCm, CUDA, MI300X

- Built an **enterprise CUDA-to-HIP migration pipeline** for AMD Developer Hackathon, coupling deterministic **AST risk scanners** (wavefront-64, PTX, WMMA) with a **MoA dual-agent** translation engine via Fireworks AI.
- Shipped **live MI300X telemetry** and readiness scoring on Vercel, reducing manual audit overhead by ∼**80%**.

OmniPitch 2026 — GitHub · Live Demo
Vue 3, Three.js, TypeScript, Tailwind CSS, Fireworks AI, Supabase, Vite

- Architected a **GenAI 3D Digital Twin** command center for FIFA 2026 stadiums using **Three.js + Vue 3** for spatial simulation, incident handling, and AI match-scenario evaluation.
- Achieved **96% line coverage** with **281 passing tests** in a production-optimized 630 KB bundle.

LearnFlow LMS — GitHub · Live Demo
Next.js, TypeScript, Django, Tailwind CSS, Fireworks AI, JWT

- Developed a **full-stack AI LMS** with dual Instructor/Student portals, real-time enrollment analytics, and **JWT auth** across **Django REST + Next.js**.
- Built the **"Test Your Might" AI Gauntlet** for on-demand assessment generation with human-in-the-loop review, automated certification, and **i18n** support.

EDUCATION

Chhattisgarh Swami Vivekananda Technical University — Bhilai, CG
B.Tech in Computer Science | ***CPI: 7.84*** — *Jul 2017 – Jun 2021*

Jawahar Navodaya Vidyalaya — Dongargarh, CG
CBSE 12th | ***GPA: 7.24*** — *Jul 2015 – Jul 2016*

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Microsoft AI Skills Fest (2026) | Google AI Essentials | Google Data Analytics

Achievements: TCS NQT 2022 — Cognitive **78.25%**, Advanced Cognitive **69.91%**, Programming **71.35%**